# ITI – PLSQL Lab 5

## Mohammad Tarek Omar – Data Management

1. Create a PL/SQL block to give a 5% bonus to all employees in the 'Sales' department who have a salary greater than 8000.

```
set serveroutput on
select * from employees;
declare

   cursor emp_cursor is
       select employees.*, department_name
       from employees, departments
       where employees.department_id = departments.department_id
       and DEPARTMENTS.DEPARTMENT_NAME = 'Salses' and EMPLOYEES.SALARY > 8000
       order by employee_id desc
       For Update;
    v_annual_salary number(8, 2);
    v_bonus number(2, 2) := 0.05;

begin

   For v_emp_record in emp_cursor Loop
    v_annual_salary := v_emp_record.salary * v_bonus;

      update employees
      set salary = v_emp_record.salary + salary
      where employee_id = v_emp_record.employee_id;
   end Loop;
end;
select * from employees;
```

2. Create a PL/SQL block to calculate the average salary in each department and print it out.

```
set serveroutput on
declare

   cursor emp_cursor is
       select departments.department_name, round(avg(employees.salary), 2) as avg_salary_per_dep
       from employees, departments
       where employees.department_id = departments.department_id
       group by department_name;

begin

   For v_emp_record in emp_cursor Loop
        dbms_output.put_line('Department name = '||v_emp_record.DEPARTMENT_NAME ||' Average Salary = '||v_emp_record.avg_salary_per_dep);
   end Loop;
end;
```

Script Output

Data Grid | Script Output

Output | Environment

```
Department name = Public Relations Average Salary = 10000
Department name = Executive Average Salary = 19333.33
Department name = Shipping Average Salary = 3475.56
Department name = Sales Average Salary = 8955.88
Department name = Finance Average Salary = 8601.33
Department name = Marketing Average Salary = 9500
PL/SQL procedure successfully completed.
```

3. Create a PL/SQL block to award a 500 bonus to all employees who have been working for more than 15years and have no commission.

```oracle
-- 3.   Create a PL/SQL block to award a 500 bonus to all employees who have been working for more than 15 years and have no commission.

set serveroutput on
select * from employees;
declare

   cursor emp_cursor is
      select employees.*
      from employees
      where EMPLOYEES.COMMISSION_PCT is null and trunc(months_between(sysdate, hire_date) / 12) > 15
      order by employee_id desc
      For Update;
begin
   For v_emp_record in emp_cursor Loop
     update employees
     set salary = v_emp_record.salary + 500
     where employee_id = v_emp_record.employee_id;
   end Loop;
end;
select * from employees;
```

Output | Grid 1 | Grid 2 | Environment

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 100 | Steven | King | SKING | 515.123.4567 | 17-JUN-03 | AD_PRES | 24000 | | | 90 |
| 101 | Neena | Kochhar | NKOCHHAR | 515.123.4568 | 21-SEP-05 | AD_VP | 17000 | | 100 | 90 |
| 102 | Lex | De Haan | LDEHAAN | 515.123.4569 | 13-JAN-01 | AD_VP | 17000 | | 100 | 90 |
| 103 | Alexander | Hunold | AHUNOLD | 590.423.4567 | 03-JAN-06 | IT_PROG | 9000 | | 102 | 60 |
| 104 | Bruce | Ernst | BERNST | 590.423.4568 | 21-MAY-07 | IT_PROG | 6000 | | 103 | 60 |

Output | Grid 1 | Grid 2 | Environment

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 100 | Steven | King | SKING | 515.123.4567 | 17-JUN-03 | AD_PRES | 24500 | | | 90 |
| 101 | Neena | Kochhar | NKOCHHAR | 515.123.4568 | 21-SEP-05 | AD_VP | 17500 | | 100 | 90 |
| 102 | Lex | De Haan | LDEHAAN | 515.123.4569 | 13-JAN-01 | AD_VP | 17500 | | 100 | 90 |
| 103 | Alexander | Hunold | AHUNOLD | 590.423.4567 | 03-JAN-06 | IT_PROG | 9500 | | 102 | 60 |
| 104 | Bruce | Ernst | BERNST | 590.423.4568 | 21-MAY-07 | IT_PROG | 6500 | | 103 | 60 |

4. Create a PL/SQL block that prints employees' first name, salary, and salary grade based on this range: for all employees

- SALARY < 5000: Grade A
- 5000 <= SALARY < 10000: Grade B
- SALARY >= 10000: Grade C

```
set serveroutput on
declare

   cursor emp_cursor is
       select employees.*
       from employees;
       v_grade varchar2(10);
begin
   For v_emp_record in emp_cursor Loop
      if v_emp_record.salary < 5000 then
         v_grade := 'A';
      elsif v_emp_record.salary < 10000 then
         v_grade := 'B';
      else
         v_grade := 'C';
   end if;
   dbms_output.put_line('First name = '||v_emp_record.first_name||' Salary = '||v_emp_record.salary||' Salary grade  = '||v_grade);
   end Loop;
end;
```

Output | Environment

```
First name = Randall Salary = 2500 Salary grade  = A
First name = Sarah Salary = 4000 Salary grade  = A
First name = Britney Salary = 3900 Salary grade  = A
First name = Samuel Salary = 3200 Salary grade  = A
First name = Vance Salary = 2800 Salary grade  = A
First name = Alana Salary = 3100 Salary grade  = A
First name = Kevin Salary = 3000 Salary grade  = A
First name = Donald Salary = 2600 Salary grade  = A
First name = Douglas Salary = 2600 Salary grade  = A
First name = Jennifer Salary = 4400 Salary grade  = A
First name = Michael Salary = 13000 Salary grade  = C
First name = Pat Salary = 6000 Salary grade  = B
First name = Susan Salary = 6500 Salary grade  = B
First name = Hermann Salary = 10000 Salary grade  = C
First name = Shelley Salary = 12008 Salary grade  = C
First name = William Salary = 8300 Salary grade  = B
PL/SQL procedure successfully completed.
```

5. Create a PL/SQL block that gives a 20% salary bonus to employees in the 'IT' department if their hire date is before 2015.

```
set serveroutput on
select * from employees;
declare
   cursor emp_cursor is
       select employees.*
       from employees, departments
       where employees.department_id = departments.department_id
       and departments.department_name = 'IT' and to_char(employees.hire_date, 'yyyy') < 2015
       For Update;
begin
   For v_emp_record in emp_cursor Loop
     update employees
     set salary = v_emp_record.salary + v_emp_record.salary * 0.2
     where employee_id = v_emp_record.employee_id;
   end Loop;
end;
select * from employees;
```

Output | Grid 1 | Grid 2 | Environment

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 103 | Alexander | Hunold | AHUNOLD | 590.423.4567 | 03-JAN-06 | IT_PROG | 9000 | | 102 | 60 |
| 104 | Bruce | Ernst | BERNST | 590.423.4568 | 21-MAY-07 | IT_PROG | 6000 | | 103 | 60 |
| 105 | David | Austin | DAUSTIN | 590.423.4569 | 25-JUN-05 | IT_PROG | 4800 | | 103 | 60 |
| 106 | Valli | Pataballa | VPATABAL | 590.423.4560 | 05-FEB-06 | IT_PROG | 4800 | | 103 | 60 |
| 107 | Diana | Lorentz | DLORENTZ | 590.423.5567 | 07-FEB-07 | IT_PROG | 4200 | | 103 | 60 |

Output | Grid 1 | Grid 2 | Environment

| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER | HIRE_DATE | JOB_ID | SALARY | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
|---|---|---|---|---|---|---|---|---|---|---|
| 103 | Alexander | Hunold | AHUNOLD | 590.423.4567 | 03-JAN-06 | IT_PROG | 10800 | | 102 | 60 |
| 104 | Bruce | Ernst | BERNST | 590.423.4568 | 21-MAY-07 | IT_PROG | 7200 | | 103 | 60 |
| 105 | David | Austin | DAUSTIN | 590.423.4569 | 25-JUN-05 | IT_PROG | 5760 | | 103 | 60 |
| 106 | Valli | Pataballa | VPATABAL | 590.423.4560 | 05-FEB-06 | IT_PROG | 5760 | | 103 | 60 |
| 107 | Diana | Lorentz | DLORENTZ | 590.423.5567 | 07-FEB-07 | IT_PROG | 5040 | | 103 | 60 |